

Zhenshuo Zhang

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Research Interests

Training algorithms in machine learning, data attribution, reinforcement learning, multitask learning, and optimization for ML.

Education

Northeastern University

Ph.D. in Computer Science

Sep 2024 – Present

Advisor: Hongyang Zhang

Zhejiang University

M.S. in Computer science

Sep 2021 – Mar 2024

Advisor: Siliang Tang

Zhejiang University

B.Eng. in Computer Science

Sep 2017 – Jun 2021

Selected Publications

Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution

Zhenshuo Zhang, Ruoxuan Xiong, Hongyang R. Zhang

Neurips 2026

- We theoretically characterize how conflicts among objectives increase regret and reduce learning efficiency in multi-objective reinforcement learning.
- We introduce routed ensemble learning to mitigate negative transfer and a trajectory attribution method to quantify nonlinear interactions across objectives.

Designing Kernel Surrogate Models for Multimodal Attribution

Ziniu Zhang*, **Zhenshuo Zhang***, Jianglin Lu, Ruoxuan Xiong, Yun Fu, Hongyang R. Zhang

Neurips 2026

- We introduce cross-sample modality attribution to quantify how a modality in one training sample affects the learning of modalities in other samples.
- Building on this formulation, we propose a structured kernel surrogate that captures nonlinear interactions across samples and modalities for attribution and data selection.

Efficient Estimation of Kernel Surrogate Models for Task Attribution

Zhenshuo Zhang, Minxuan Duan, Hongyang R. Zhang

ICLR 2026

- We theoretically prove that influence functions and datamodels are equivalent: both provide first-order estimates of data attribution.
- Building on this analysis, we introduce a kernel surrogate model that captures nonlinear interactions among training data and tasks.

Scalable Multi-Objective and Meta Reinforcement Learning via Gradient Estimation

Zhenshuo Zhang, Minxuan Duan, Youran Ye, Hongyang R. Zhang

AAAI 2026

- We use a first-order gradient estimation of PPO outcomes to efficiently predict policy performance when training on different combinations of tasks.
- Using the predicted performance, we optimize a partition of the task set and train an ensemble of policies, one for each task group.

Linear-Time Demonstration Selection for In-Context Learning via Gradient Estimation

Ziniu Zhang*, **Zhenshuo Zhang***, Dongyue Li, Lu Wang, Jennifer Dy, Hongyang R. Zhang

*EMNLP 2025; * equal contribution*

- We use a first-order gradient estimation to efficiently predict LLM output quality under different combinations of prompts.
- Based on the predicted LLM performance, we select the best-performing sequence of demonstrations.

Other Publications

WinQ: Accelerating Quantization-Aware Training of Language Models Around Saddle Points

Dongyue Li, Zechun Liu, Kai Yi, **Zhenshuo Zhang**, Changsheng Zhao, Raghuraman Krishnamoorthi, Harshit Khaitan, Hongyang R. Zhang, Steven Li

ICML 2026

Efficiently Learning Branching Networks for Multitask Algorithmic Reasoning

Dongyue Li, **Zhenshuo Zhang**, Minxuan Duan, Edgar Dobriban, Hongyang R. Zhang

KDD 2026

One-Sided Matrix Completion from Ultra-Sparse Samples

Hongyang R. Zhang, **Zhenshuo Zhang**, Huy L. Nguyen, Guanghui Lan

Transactions on Machine Learning Research

GraphControl: Adding Conditional Control to Universal Graph Pre-trained Models for Graph Domain Transfer Learning

Yun Zhu, Yaoke Wang, Haizhou Shi, **Zhenshuo Zhang**, Dian Jiao, Siliang Tang

WWW 2024

MARIO: Model Agnostic Recipe for Improving OOD Generalization of Graph Contrastive Learning

Yun Zhu, Haizhou Shi, **Zhenshuo Zhang**, Siliang Tang

WWW 2024

Structure-Aware Group Discrimination with Adaptive-View Graph Encoder: A Fast Graph Contrastive Learning Framework

Zhenshuo Zhang, Yun Zhu, Haizhou Shi, Siliang Tang

ECAI 2023

Talks

- Gave a talk at INFORMS (Oct 2025).

Honors and Awards

- Zhejiang Provincial Outstanding Graduate Award, 2024
- Outstanding Undergraduate Award, Zhejiang University, 2021